MATERNAL HEALTH
AND SAFE MOTHERHOOD PROGRAMMES

PREVENTION AND MANAGEMENT OF SEVERE ANAEMIA IN PREGNANCY

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1. PREFACE

About half a million women die each year following complications of pregnancy and childbirth, and many times that number suffer ill-health or disability.[1] Many of these deaths could be prevented through properly organized primary health care and accessible referral facilities.

The great majority (99%) of maternal deaths occur in developing countries. In response to the enormity of the problem, the Safe Motherhood Initiative (SMI), an inter-agency effort to reduce maternal mortality and morbidity, was launched in 1987. Its target is to reduce levels of maternal deaths by at least half by the year 2000 and to achieve substantial reductions in maternal morbidity.

Most of the necessary knowledge and technology to achieve this reduction are already available. What is missing in many of the developing countries is the ability to put this knowledge into practice. In order to be able to provide more effective support to developing countries, WHO has been holding a series of meetings and consultations with experts on a variety of topics relating to maternal health. The aim is to review current knowledge and experience on high priority topics, produce managerial and clinical guidelines, and recommend needed epidemiological and operational research. Reports of such meetings include *Essential obstetric functions at first referral level*,[2] *Prevention and treatment of obstetric fistulae*,[3] *Prevention and management of postpartum haemorrhage*,[4] and *Detecting pre-eclampsia*.[5]

This meeting of the Technical Working Group on the prevention and treatment of severe anaemia in pregnancy is part of the effort by WHO to provide more effective support to developing countries in the development of national Safe Motherhood programmes.

2. INTRODUCTION

Anaemia directly or indirectly contributes to a significant proportion of maternal deaths in the developing world (see Table 1). Severe anaemia can lead to cardiac failure in pregnancy, while lesser grades of severity are associated with decreased maternal well-being and contribute to maternal deaths from haemorrhage or infection.[6] Anaemia may also contribute to perinatal morbidity and mortality by increasing the likelihood of intra-uterine growth retardation[7] and preterm delivery.[8]

The effects of anaemia on maternal and perinatal morbidity and mortality are largely preventable with appropriate treatment. Although the epidemiology of anaemia, and the knowledge and technical means of preventing and controlling it are well known, there are as yet few MCH programmes that successfully apply this knowledge to the development and implementation of comprehensive control strategies in developing countries.

A WHO meeting of experts with relevant experience in this area was therefore convened in Geneva to address this issue and propose appropriate recommendations for action. The list of participants of this Technical Working Group is in Annex 2. The background paper for the meeting was prepared by Dr Andrew Hughes.

The objectives of this meeting were to:

(a) Review the reasons why anaemia in pregnancy continues to be a widespread problem despite the present state of knowledge on its aetiology and modes of treatment.

(b) Suggest strategies for overcoming current constraints of anaemia prevention as part of MCH/FP programmes.

(c) Draft practical guidelines for health workers and those responsible for the planning and management of MCH/FP services on the prevention and management of anaemia in pregnancy at different levels of care.

(d) Identify gaps in present-day knowledge and suggest research priorities.

Table 1: Contribution of anaemia to maternal mortality[9]

	Anaemia deaths as % of all maternal deaths	Anaemia deaths per 100 000 live births
Kwale District, Kenya*	11	82
Malawi**	8	48
Ilesha, Nigeria**	9	83
Dakar, Senegal	5	35
Tangail District, Bangladesh*	4	47
Thimpu, Bhutan**	7	55
Andhra Pradesh, India*	9	27
Ambala, Northern India*	16	38
Karachi, Pakistan**	7	194

* Community study
** Hospital study

3. DEFINITIONS

The WHO definition of anaemia in pregnancy[10] of a haemoglobin concentration of less than 110 g/l was accepted for the purpose of the meeting. However, in the second trimester of pregnancy, a fall of 5 g/l due to the increase in plasma volume should be allowed[11] and a cut-off value of 105 g/l used.

The following definition of degrees of severity, as recommended in the Report of the African Regional Consultation on Control of Anaemia in Pregnancy, in Brazzaville,[12] was adopted:

Degree of anaemia	Haemoglobin (Hb) (g/l)	Haematocrit (PCV) (%)
Moderate	70-109	24-37
Severe	40-69	13-23
Very severe	<40	<13

4. PREVALENCE OF ANAEMIA IN PREGNANCY

A recent update of the WHO database[13] on the prevalence of nutritional anaemia in women indicates that about half of the pregnant and a third of the non-pregnant women in the world suffer from nutritional anaemia. A comparison between developed and developing countries shows that while in the former approximately 10% non-pregnant and 15% pregnant women are anaemic (Hb <110 g/l), in developing countries the figures are 42% and 56% respectively. The situation is particularly severe in Southern Asia where three quarters of pregnant women are anaemic (see Table 2).

Table 2: Estimated prevalence of mild, moderate and severe anaemia in pregnant women (around 1988)

	Hb <70 g/l %	Hb 70-99 g/l %	Hb 100-109 g/l %	No. of countries	No. of subjects
AFRICA					5,687
Eastern	4	27	16	6	4,632
Middle	5	28	21	3	222
Northern	3	25	25	1	1,936
Southern	2	18	15	1	22,131
Western	4	33	19	6	
ASIA*					
Eastern	ǂ	24	13	ǂ	ǂ
South-eastern	5	39	19	4	15,811
Southern	7	43	25	4	14,347
Western	2	28	20	4	3,363
LATIN AMERICA	2	33	17	12	11,058
Caribbean	2	26	14	2	841
Central	1	24	12	4	7,261
Southern					
OCEANIA**	3	42	26	6	4,211

ǂ Unknown
* Excluding Japan
** Excluding Australia and New Zealand

Source: WHO database: unpublished estimates

At the African Regional Consultation on Control of Anaemia in Pregnancy,[12] it was decided to classify the prevalence of anaemia as high, moderate or low, according to the following categories:

Prevalence	Degree of anaemia	
	Anaemia (70-109 g/l)	Anaemia (Hb <70 g/l)
High	≥40%	≥10%
Medium	10-39%	1-9%
Low	1-9%	0.1-0.9%

This classification does not suggest that action is necessary only where the prevalence of anaemia is high. But these categories are felt to be useful for comparative purposes.

5. CAUSES OF ANAEMIA IN PREGNANCY

Anaemia in pregnancy is often of multiple aetiology. Iron and folate deficiency is by far the most important aetiological factor,[14] and is aggravated by short birth intervals and parasitic and helminthic infections. Malaria, causing the destruction of red blood cells, contributes significantly to the prevalence of anaemia in a population and is often associated with iron and/or folate deficiency. AIDS should be considered in the differential diagnosis of a patient with anaemia. Haemoglobinopathies such as sickle cell disease contribute to the causation of anaemia in Africa.

In normal pregnancy, there is a fall of about 5 g/l in the haemoglobin concentration. This is called the "physiological anaemia of pregnancy" and is due to the disproportionate increase in plasma volume with respect to the red cell mass.[15] This process occurs mainly in the second and third trimesters and is not pathologic in itself, but could aggravate any anaemia already present.

5.1 Iron deficiency

Iron requirement is increased in pregnancy, due to the demands of the fetus and the increase in blood volume, especially in the last two trimesters, with up to 80% of the requirement relating to the last trimester.[16] The total iron requirement over the whole pregnancy is about 1000 mg (300 mg for the fetus, 50 mg for the placenta, 450 mg for the increase in the maternal red cell mass and 240 mg for the continuing maternal basal iron losses). Requirements during the first trimester are relatively small, 0.8 mg per day, but rise considerably during the second and third trimesters to a high of 6.3 mg per day. After delivery and during lactation, iron requirements decrease to 1.31 mg per day which is less than the requirements in menstruating women (2.38 mg/day).[17] Despite an increase in iron absorption in pregnancy, diet alone is unable to satisfy the increased requirement. Therefore the extra requirement for iron has to be met by the body's iron stores. However, iron stores are low or absent in the great majority of women in developing countries due to a diet low in iron, chronic blood loss from parasitic infections, and frequent and closely spaced pregnancies. (It has been estimated that in the absence of

iron supplementation, it can take up to two years to regain pre-pregnancy iron status.[18]) Iron stores of women during reproductive years are further depleted by loss of blood from menstruation.

Thus, while in the industrial world a significant proportion of women of reproductive age may be iron deficient, though probably not anaemic, in many developing countries the prevalence of both anaemia and iron deficiency is much greater. The prevalence of iron deficiency may be 2-3 times that of actual anaemia, ranging from about 50% in some countries to nearly 100% in parts of others.[14]

Iron deficiency is mainly nutritional in origin. There are two types of dietary iron: haem and non-haem. Haem iron is found in foods of animal origin such as meat, fish, poultry and its bioavailability is high with absorption being 20-30%. Non-haem iron is found in foods of plant origin, especially whole grain cereals, tubers, vegetables and pulses. Its bioavailability is lower and is determined by the presence of enhancing and inhibiting factors consumed in the same meal.

Enhancers of non-haem iron absorption include meat, poultry, fish and vitamin C. Meat, poultry and fish are therefore of double value in that not only do they provide a rich source of bioavailable iron, but also enhance the absorption of non-haem iron contained in the rest of the meal. In developing countries, where meat intake is low, vitamin C is the most important enhancer of iron absorption. The addition of as little as 50 mg of vitamin C to a meal can double iron absorption.[19] This could be provided by an orange, 120 g of papaya or mango, or 100 g of raw cabbage.

Phytates and polyphenols (which include tannins) inhibit iron absorption. Phytates are present in wheat and other cereals, and even small amounts can markedly reduce iron absorption. Tannins, present in tea and to a lesser extent, coffee, and other polyphenols found in nuts and pulses also inhibit iron absorption. The inhibitory effects of phytates and polyphenols can be counteracted by adding vitamin C to a meal.

In addition to iron derived from food, the diet may also contain exogenous iron originating from the soil, dust, water or cooking vessels. In developing countries, cooking foods in iron pots, especially vegetable soups of low pH, can considerably increase the iron content of a meal.

As compared to diets in industrialized countries, diets in developing countries contain much fewer foods of animal origin and fewer foods rich in vitamin C, thus decreasing the bioavailability of iron. In many developing countries, meals are based on cereals, roots and tubers and contain negligible quantities of meat, fish or vitamin C and a preponderance of foods that inhibit iron absorption such as maize, rice, beans, wholewheat flour and sorghum.

Contributing to dietary iron deficiency in developing countries is the iron deficiency caused by infections. Infections causing chronic blood loss such as parasitic infestation with hookworm and to a lesser extent schistosoma, increase iron requirements. Viral and bacterial infections interfere with food intake, absorption, storage and use of many nutrients including iron. Repeated episodes of infection may thus contribute to the development of iron deficiency and anaemia.

Apart from causing depleted iron stores and anaemia, iron deficiency itself can have an independent effect on maternal health. On the one hand, it has been suggested that

iron deficiency hinders the immune system's ability to fight infection.[20] On the other, it is reported that both bacterial infections and malaria are less common in iron deficiency states.[21] In support of the latter hypothesis, a study from Papua New Guinea reports that giving iron infusions to pregnant women was associated with an increased incidence of blood films positive for malaria in primigravidae.[22]

5.2 Hookworm infection

Hookworm infection is one of the principal causes of iron deficiency anaemia in developing countries. About one quarter of the world's population has hookworm infection. It is prevalent throughout the tropics and subtropics wherever there is faecal contamination of the environment and is acquired mainly by skin contact with contaminated soil or vegetation.

Adult hookworms live in the duodenum and jejunum, attached to the intestinal mucosa from which they suck blood, causing a chronic blood loss. Over a period of time, even small hookworm loads may cause sufficient blood loss to deplete body iron reserves. In people whose dietary intake of iron is low and whose body iron stores are already depleted, hookworm infection can give rise to iron deficiency anaemia in just a few weeks, especially during pregnancy, when iron requirements are increased.

5.3 Folate deficiency

Folates are a heat-labile, light sensitive family of water-soluble vitamins essential to red blood cell maturation. Folate deficiency produces an anaemia characterized by unusually large red blood cells (megaloblastic anaemia). Folates are present in almost all foods, but more plentiful in liver, dark green vegetable leaves, yams, sweet potatoes, egg yolk, fish, pulses, nuts and fruits such as bananas, plantains, and mangoes. Some important staples in the developing world such as rice, cassava, millet, sorghum and maize, are poor sources of folates.

The bioavailability of folates from foods is a less important factor in the causation of anaemia than iron deficiency. More important is the fact that folates are heat labile. This means that prolonged cooking and repeated reheating of food, which is a common practice in the developing world, can be an important factor in the aetiology of folate deficient anaemia.[23]

Folate requirements approximately double during pregnancy, especially in the last trimester and in the puerperium. Since body stores of folate are limited and dietary folate is likely to be insufficient in developing countries, anaemia often develops as a consequence. Many studies have demonstrated a steady fall in serum folate levels throughout pregnancy, especially in women from lower socioeconomic groups, in multigravidae, in smokers, and in women with twin pregnancies.[24]

The requirement for folate is also increased by disease processes associated with haemolysis such as malaria or sickle cell disease. Thus malaria and folate deficiency often coexist in pregnant women. In one study in West Africa, antimalarial prophylaxis alone, without folate supplements, reduced the frequency of megaloblastosis in primigravidae by 50%, and the addition of folate supplementation completely abolished it.[25]

Limitation of growth and pelvic development has been observed in folate deficient pregnant girls below the age of 16 years in Nigeria.[26] When compared with girls who did not receive supplements, a significantly greater proportion of girls who were supplemented grew by ≥2 cm during pregnancy. There was an associated development of the pelvis and a highly significant reduction in the number of operative deliveries for disproportion.

Iron and folate deficiency, both being of nutritional origin, tend to coexist in the same subject. This point should be kept in mind, as iron deficiency may conceal the presence of a coexisting megaloblastic process.

There is good evidence that folate supplements given to the mother around the time of conception reduces the frequency of neural tube defects in infants.

5.4 Malaria

Malaria is a major factor causing severe anaemia in pregnancy. Anaemia associated with malaria is caused by haemolysis of the red blood cells. Hypersplenism, a condition characterized by exaggeration of the inhibitory or destructive functions of the spleen, contributes to the anaemia in up to 25% of women[27] who suffer from malaria in pregnancy.

Longitudinal studies have shown that parasite infection rates are greater in pregnant women as compared to non-pregnant women (especially in primigravidae), and that parasitaemia could be abolished in pregnant women receiving chemoprophylaxis.[25,28,29,54] Several studies have shown that protection against malaria contributes to the prevention of anaemia in pregnancy[25,28,54] thus highlighting the importance of chemoprophylaxis and other methods of malaria control in reducing maternal morbidity and mortality.

Malaria is associated with an increased incidence of low birth weight,[30,31] premature deliveries[32] and stillbirths.[33,34] The adverse effects of malaria on maternal and fetal well-being are for the most part due to the associated severe anaemia. However, a direct effect of malaria itself cannot be ruled out.

5.5 Sickle cell disease (SCD)

The sickle cell trait is carried by about 60 million people in the world, about 50 million of whom are in Africa. Persons with sickle cell trait (heterozygotes) have minimal clinical problems but homozygotes have sickle cell disease (SCD), which causes chronic haemolytic anaemia. Over 100 000 infants are born each year with SCD, most of them in Africa. Anaemia is usually severe and may be exacerbated by acute sequestration of sickled cells in the spleen or more commonly by the "aplastic crisis", which occurs when bone marrow red blood cell production slows down during acute infections. Folic acid and iron deficiency are often associated with SCD because the haemolytic process increases the requirement for these nutrients.

Sickle cell disease is currently not an important factor in the actiology of anaemia in pregnancy because few girls with SCD currently survive to reproductive age. However, with the improvement of social conditions and the establishment of sickle cell clinics in Africa, survival will improve and SCD will contribute more and more to the problem of anaemia in pregnancy. There are also a number of clinically milder sickling disorders associated with the interaction of haemoglobin S (HbS) with other abnormal

haemoglobins, such as haemoglobin C (HbC) or with the alpha or beta thalassaemias. Girls affected with these conditions do survive into adult life.

5.6 Human immunodeficiency virus (HIV)

Because it is becoming increasingly common in sexually active men and women, especially in Africa, where reported frequencies of seropositivity in antenatal clinics range from 6 to 24%,[35] HIV infection must now be included in the differential diagnosis of anaemia in pregnancy. When anaemia is associated with leucopenia and thrombocytopenia, the antenatal health worker should be alerted to the possibility of AIDS, especially in a woman not responding to therapy.

Transmission of HIV infection by blood transfusion is common in developing countries, where there is a relatively high prevalence of HIV positivity among donors and where the ability to screen for HIV is suboptimal. This further highlights the importance of the antenatal clinic in the prevention of anaemia early in pregnancy, which may avoid the need of a blood transfusion later in the pregnancy.

6. DIAGNOSIS AND ASSESSMENT OF ANAEMIA IN PREGNANCY

The successful management of anaemia in pregnancy depends on reliable techniques for detecting anaemia, assessing its severity and monitoring the response to appropriate treatment. In the industrial world this is readily accomplished by measurement of the haemoglobin concentration (Hb) using an electronic counter. In the developing world, however, such sophisticated equipment is largely inappropriate due to the excessive capital expense and running costs, lack of suitably trained laboratory technologists, difficulty with regular maintenance and an electricity supply that is often unstable, intermittent or absent. This is particularly true in rural areas where the prevalence of anaemia is most likely to be highest and where appropriate prevention and treatment strategies may have the most beneficial effects. Even in urban areas the lack of a dedicated laboratory and trained technologists may further limit the techniques available for anaemia detection and assessment.

Ideally, laboratory tests to detect anaemia should be carried out periodically throughout a pregnancy. This is best done either by measuring Hb concentration or the haematocrit/packed cell volume (HCT/PCV). However, even relatively simple laboratory techniques may not be available in many rural areas. In these circumstances an alternative strategy is to use a clinical or simple screening test for anaemia on all pregnant women and to reserve the more technically demanding laboratory measurements for those women who are judged to be significantly anaemic on screening. Thus further discussion will concentrate first on screening tests for anaemia, and second on laboratory techniques for the more accurate assessment of the degree of anaemia. The former techniques are particularly useful for defining those at high risk, while the latter are essential for the proper assessment of the severity of anaemia and in the monitoring of responses to therapy.

There are two major types of diagnostic techniques:

1. Screening tests, which indicate whether the haemoglobin level is above or below a given cut-off value.
2. Tests to assess the degree of anaemia: these examinations require equipment that will allow the actual measurement of the haemoglobin concentration.

6.1 Screening tests

6.1.1 Clinical screening for anaemia (qualitative)

Studies using subjective interpretation of paleness of conjunctivae, skin and nail beds[36,37] as an estimate of anaemia have shown low sensitivities (38% and 24%, respectively), but reasonable specificities (90% and 68%, respectively). A recent as yet unpublished study from the Gambia[38] compared conjunctival paleness with colorimetric haemoglobin determinations in 540 antenatal pregnant women in two rural clinics. Sensitivity was poor at 16%, but specificity was excellent at 100%. Sensitivity was, however, much improved when Hb concentrations were below 70 g/l.

In an attempt to improve accuracy of screening for anaemia, a conjunctival recognition card was produced by some health groups and governments, but the problem of low sensitivity remained. Sanchez-Carrillo et al[39] have also reported their experiences with the use of a non-invasive instrument to make the assessment of conjunctival colour more accurate. The instrument consists of six photographic images of eye conjunctivae with different shade intensities, corresponding to Hb levels between 70-200 g/l. Sensitivity was 58% for Hb between 70-110 g/l, with specificity of 86%. A similar non-invasive instrument has also been developed for comparing the colour of the tip of the tongue with that of a prepared colour scale of increasing intensity of colour shades.[9] It is still being evaluated.

6.1.2 Other screening for anaemia (qualitative)/(cut-off values)

Other screening methods include the Tallqvist technique and the use of copper sulphate solutions. They are all invasive techniques, involving a finger prick to produce a drop of blood for use in the test.

The Tallqvist technique

As a method of estimating haemoglobin the Tallqvist technique is the simplest and cheapest to use. The method is based on the direct visual comparison of a drop of whole blood on a filter paper against a standard scale of colours equivalent to haemoglobin concentrations ranging from 30 to 100%. The test should be done in daylight, and as quickly as possible after the blood sample has dried.

The copper sulphate method

This test is based on the fact that the Hb concentration of a blood sample can be estimated from its specific gravity.[40] A copper sulphate solution is prepared with a specific gravity equivalent to a specific Hb value. A drop of capillary blood is allowed to fall into the solution and if it continues to fall, its specific gravity is

greater than that of the reference haemoglobin value. Should the drop rise to the surface, then its specific gravity and its corresponding Hb concentration is less than that of the solution. This simple screening test is used for donors in the British and Swiss Blood Transfusion Services. The copper sulphate method also has an educational value in that women can observe the "heaviness" or "thickness" of their own blood.

There have been several studies evaluating the above screening methods.[41,42] Although clinical examination of the conjunctiva, or other areas of the body is not very sensitive, it has the advantage of being both simple and non-invasive. Thus, in the absence of any other more sensitive technique, especially in rural areas, antenatal clinic workers could be taught and encouraged to use this rapid and simple screening method. The Tallqvist technique is more reliable than clinical assessment, but appears to be less accurate than the copper sulphate technique. The latter is probably the screening method of choice and certainly lacks the subjectivity of the other two.

6.2 Methods for assessing the degree of anaemia

6.2.1 Haematocrit/packed cell volume

This is technically simple and measurements can be done on site or samples may be collected in heparinized capillary tubes for later estimation. However, the major disadvantage is the requirement for a haematocrit centrifuge, which is an expensive item and usually requires a source of electricity, although some battery operated centrifuges are available. Packed cell volume measurement is also best suited to batching samples and is more tedious for individual measurements or small numbers of samples.

6.2.2 Haemoglobinometers (comparators) - the AO Spencer/BMS haemoglobinometer

A drop of undiluted whole blood is placed in a glass chamber and lysed with a saponin impregnated stick. The chamber is then placed in the instrument in front of a battery-lit bulb emitting green light. The intensity of the colour is then compared with a graded standard on a split screen. This instrument has been shown to be well-liked by health workers, it is easy to use and is also accurate and sturdy under field conditions.[41]

6.2.3 Haemoglobinometer (digital) - the "HemoCue"

The "HemoCue" measures Hb at two wavelengths as azide methaemoglobin. It uses undiluted blood samples that are lysed and the haemoglobin level is displayed digitally in g/l. It has excellent accuracy. One type that is available is dependent on disposable cuvettes, while another type under development would employ reusable cuvettes. The latter appears very suitable for developing countries.

6.2.4 Dilution techniques

The dilution methods of haemoglobin measurement require an accurate measurement of a volume of blood. While this poses little problem in industrial countries where dilution is usually automated, it is a major problem in developing countries where it is largely a manual procedure. These methods can lead to inaccuracies in haemoglobin measurement and are not recommended for general use in developing countries.

A review of the available methods for assessing the degree of anaemia has been published.[43]

6.3 Diagnosis of the cause of anaemia

6.3.1 Iron and folate deficiency

Severe iron deficiency causes hypochromia and microcytosis of the red blood cells, which can be readily recognised on a stained thin blood film. Moderate iron deficiency produces less marked changes of hypochromia and microcytosis and in mild deficiency, the blood film may look normal. Here, additional laboratory investigations may be necessary to confirm iron deficiency. These include the serum iron (SI), the serum ferritin (SF), the iron binding capacity (IBC), the transferrin saturation (TS), and the bone marrow aspirate, which is examined for the presence of stainable iron in order to estimate iron stores. In iron deficiency, a decrease of stainable bone marrow iron and a decrease in the serum ferritin are the first detectable changes. This is followed by a stage of iron deficient erythropoiesis where the SI falls to less than 10 µmols/l and the TS decreases to less than 15%. Stainable bone marrow iron is absent and there is a further decrease in SF to less than 10 µg/l. In both these stages the Hb concentration is normal and the thin blood film is normal. Finally, if negative iron balance persists, a microcytic, hypochromic anaemia develops associated with further falls in SF, SI and TS.

Folate deficiency causes macrocytosis of the red blood cells and hypersegmentation of neutrophils, apparent on a thin blood film. There are in addition megaloblastic changes, which can be recognized on bone marrow aspirates. There is also a reduction in serum and red cell folate levels. A combined iron/folate deficiency may mask the typical changes of either deficiency on thin blood film, though a double population of hypochromic/microcytic and macrocytic cells may be seen.

In areas where laboratory facilities are limited, the most useful and simplest diagnostic tool for iron and folate deficiency is the examination of a thin blood film, as long as there is a trained technician and a microscope. A particularly suitable battery-operated, binocular microscope is available for use in rural areas of the developing world.[44] The battery can be charged via a solar panel and the instrument may also operate from the electrical power supply.

6.3.2 Malaria

Malaria is diagnosed by demonstrating parasites in Giemsa-stained thick and thin blood films. *Thick films* are many cells thick and contain 6-20 times as much blood per unit area as a thin film. They are therefore suited for rapid diagnosis of parasitaemias not readily detected in thin films. *Thin films* are useful for studying the details of the parasites, and for identifying the species of parasite. When malaria is suspected and the initial films are negative, fresh blood should be examined at 6 hour intervals for 3 consecutive days.

6.3.3 Sickle cell disease

Sickle cell disease can be diagnosed by a history and a thin blood film, which shows sickle cells, target cells and reticulocytes. Haemoglobin electrophoresis will determine the type of haemoglobin present.

7. PREVENTION AND TREATMENT OF ANAEMIA IN PREGNANCY (GENERAL PRINCIPLES)

7.1 Supplementation with iron and folate

While a degree of controversy surrounds routine supplementation with iron and folate during pregnancy in countries with low prevalence of anaemia, there should be no doubt about its necessity in countries where iron and folate deficiency is more common. Adverse effects on maternal and fetal health from iron and folate deficiency can be prevented or reversed by adequate supplementation.

Studies from developing countries[45,46] suggest that 120 mg elemental iron and 1 mg folic acid are the optimum daily dosages needed to prevent anaemia in pregnant women. A suitable, recommended preparation would be a combination tablet containing 60 mg of elemental iron and 250 µg of folate, to be taken twice a day. Where this is not available, tablets such as ferrous sulphate, containing 60 mg of elemental iron, should be given twice a day together with one folic acid tablet (1 mg).

The usual daily dosage of elemental iron thought to be sufficient in pregnancy in industrialized countries is 60 mg. That it should be twice as much in the developing world is not surprising, as many more women start off pregnancy with low or absent iron stores due to poor nutrition and frequent infections, particularly hookworm infection and malaria.

When anaemia is already present, especially if severe, higher therapeutic doses of iron should be given, usually 180 mg of elemental iron and 2 mg of folic acid daily. Erythropoietic activity following iron administration is directly related to severity of anaemia, the increase in haemoglobin concentration being inversely proportional to the initial concentration. Iron absorption is therefore greatest in the first few weeks of treatment but then rapidly falls off as the haemoglobin concentration rises towards normal; it may be as low as 2% after 4 months of treatment.[47] The first month of treatment would thus appear to be the most important time for ensuring the success of treatment. A positive response to treatment can be defined as a daily increase in haemoglobin concentration of about 1 g/l/day from the fourth day on (approximately 7 g/l/week), with normal levels reached within 4 to 8 weeks depending on the original severity of the anaemia.[48] Iron treatment should continue for at least another 2 or 3 months to build up iron stores to about 250-300 mg, which is equivalent to a serum ferritin of 30 µg/l.[17]

7.2 Compliance to oral supplementation

The effectiveness of supplementation programmes is constrained by the problem of non-compliance. A woman's belief system, her perceptions and knowledge about health, greatly influence her decision to follow through with a prescribed preventive or treatment regime. For example, many pregnant women in India and Thailand[49,50] believe that taking iron and vitamin tablets will cause them to have big babies and thus difficult deliveries, and they therefore resist iron and folate supplementation. In other societies, iron is perceived as having "heating" qualities; and since pregnancy is considered to be a time of increased body heat, it is believed that taking iron pills could cause "overheating" and induce abortions.[49]

Factors such as the form in which a drug is administered, the colour and the taste also influence women's compliance with a drug therapy. For example, in some areas,

injections or liquid medicines are considered more appropriate during pregnancy than pills.[51] The task for health workers is to encourage and build on beliefs that are beneficial, to convince women to take iron supplements. For example, in Sierra Leone, red foods and medications are believed to replace or purify blood.[52] If health workers explain how anaemia in pregnancy is a disorder that affects the blood, red iron tablets then may be considered as an acceptable treatment measure.

Another important factor affecting compliance is a woman's perception of pregnancy. Pregnant women are, in many societies, expected to carry on as usual and the symptoms of tiredness and breathlessness are accepted as normal. Thus non-compliance is more of a problem among women with mild and moderate anaemia when symptoms are not very marked. Women who are severely anaemic, with severe breathlessness even at rest, notice the beneficial effects of iron supplementation very quickly and continue with the treatment.

Women in developing countries also generally lack a good understanding of the relationship between anaemia and pregnancy outcome. Explanation about the relationship of anaemia and blood loss during delivery might help to alert women to the need to take iron supplementation. It is also generally true that most women will comply with any therapy if they can be convinced that it will be good for the baby.

Side-effects of oral iron, which include epigastric discomfort, nausea, vomiting, constipation or diarrhoea, are another factor affecting compliance. The frequency of side-effects is directly related to the dose of iron and thus the amount absorbed. The most effective dose of iron in high prevalence countries, 120 mg daily, is more likely to produce side-effects than a conventional lower dose, and thus more associated with non-compliance. It is debatable whether lower doses of iron, which are likely to lead to fewer side-effects, are less effective in preventing iron deficiency. Research is presently being done to clarify this relationship.

Many health workers who prescribe iron therapy are themselves unaware that anaemia adversely affects pregnancy outcome. There is thus a need to teach health workers that anaemia prevention should be one of their priorities and to give them communication skills to enable them to persuade women to take iron supplementation. Traditional health workers such as TBAs can be valuable allies because they have detailed knowledge of local customs and beliefs and are trusted by the community.

Because of the problem of compliance with regular oral iron supplementation, parenteral iron therapy is being considered as a possible alternative measure. The most commonly used preparation is iron dextran. Administration of parenteral iron therapy, in addition to avoiding the problem of non-compliance, has the major potential benefit of providing rapid and complete correction of iron deficiency, including replacement of the iron stores, and can either be given as a series of intramuscular injections or in a single dose as an intravenous infusion (Total Dose Infusion of iron, TDI). TDI is associated with a rapid rise in haemoglobin over the first 2 weeks, with response rates varying from 10 to 20 g/l/week (double or triple the rate of rise in haemoglobin following oral iron treatment), with the highest rates seen in the most anaemic patients. These higher response rates seem to fall off after 2 weeks as the haemoglobin concentration increases, and thereafter similar response rates are seen for TDI and oral iron. The drawbacks of intramuscular injections are that they may be painful and associated with staining of the skin. TDI has been associated with severe anaphylactic reactions, especially in those with a history of allergy. The number of reactions to TDI can be decreased by premedicating

the patient with an antihistamine, giving the first 10 minutes of the infusion very slowly and not giving intravenous iron to patients with an allergic history. Local thrombophlebitic reactions may be decreased by using normal saline as the diluent and by not putting more than 30 ml of iron-dextran in one litre of solution. Parenteral iron is also relatively expensive and requires adequate facilities and supervision. Because of these disadvantages, parenteral iron therapy is not used routinely, but should be considered in cases of low tolerance of oral iron, poor compliance and severe iron deficiency in late pregnancy. The latter is a particularly good indication for TDI: because of the rapid rise in haemoglobin associated with TDI, it may decrease the need for blood transfusion for severe anaemia prior to delivery or in the postnatal period and make postpartum haemorrhage less likely to result in death.

7.3 Dietary modification

Dietary modification is an important way of decreasing the incidence of nutritional deficiency, though this may be difficult due to the limited purchasing power of a household and problems in persuading people to change long-established dietary habits. Increased consumption of normal foods so that energy needs are met can increase iron consumption. Such a strategy in rural India increased total iron consumption by 25-30%.[53,54] Enhancing the bioavailability of ingested iron rather than increasing the total amount in the diet is another possibility. This can best be achieved by promoting the intake of iron absorption enhancers such as haem iron (e.g. meat) and vitamin C. Any campaign to promote the intake of haem iron could run into the obstacle of its unavailability and high cost, as well as religious and philosophical objections to the consumption of meat. In contrast, efforts to increase the content of non-haem iron, which is found in foods such as pulses and green vegetables in conjunction with a greater vitamin C intake, have a greater chance of success. In many communities, vegetables and fruits are eaten only infrequently and in small amounts. Persuading families to add these foods to their diet can have a considerable impact on iron absorption. Pregnant women should also be encouraged to increase their intake of pulses and green vegetables. Prolonged cooking of vegetables should also be discouraged as this destroys much of the vitamin C and folic acid content; nutrition education should stress this and teach ways of avoiding it. Health workers should also stress the importance of reducing tea and coffee consumption, particularly during meals. This alone can have a considerable impact on bioavailability of iron.

7.4 Food fortification

Food fortification has the advantage of requiring little or no cooperation from the community and adds little to the price of food. A widely consumed food should be selected and fortified with iron in a form that is adequately absorbed, without altering the taste or appearance of the food. Examples of foods that have been successfully fortified include bread, milk, formula, salt, sugar, and fish sauce. Food fortification is a promising way of controlling nutritional deficiencies, but it does require an industrial infrastructure that does not exist in some developing countries.

7.5 Malaria control

Malaria may be an important contributor to severe anaemia, especially in primigravidae. Chemoprophylaxis has been shown to abolish parasitaemias and prevent anaemia, provided it is not started before the 19th week in pregnancy.[25,54]

Chemoprophylaxis also reduces the risk of megaloblastic anaemia resulting from the increased folate requirements associated with malaria-induced haemolysis.[25,54]

Malaria chemoprophylaxis is likely to be of greatest benefit in primigravidae residing in holoendemic areas; in all pregnant women during times of high transmission periods; and in those residing where seasonal patterns exist. It has been suggested that all pregnant women in endemic areas should have a curative dose of chloroquine on their first prenatal visit (600 mg base at once, 600 mg on the second day, and 300 mg on the third); followed by daily proguanil (100 mg/day, 200 mg if over 160 cm in height) for the duration of pregnancy. This was a specific recommendation for Africa and was successful when used in Northern Nigeria,[25] but may have to be amended for other regions and countries, especially where there is widespread chloroquine resistance. Other methods of malaria control should not be forgotten, such as the use of bednets impregnated with an insecticide such as permethrin. Non-immune pregnant women should be discouraged from visiting malarious endemic areas.

7.6 Control of other parasitic infections

Control and treatment of parasitic infections other than malaria is important, particularly hookworm infection and schistosomiasis (which can contribute to iron deficiency through chronic blood loss), and other parasites that interfere with the absorption of nutrients such as giardiasis.

Where women are exposed to heavy hookworm infection, albendazole or mebendazole could be given, although there is some risk to the fetus, especially in the first trimester. Routine deworming is unlikely to have a major impact unless parallel measures are taken to prevent or decrease reinfection such as the provision of cheap footwear, the provision of safe water and improvements in sanitation and personal hygiene.

8. STRATEGIES FOR OVERCOMING CURRENT CONSTRAINTS TO ANAEMIA PREVENTION PROGRAMMES

As has been demonstrated, the aetiology of anaemia in pregnancy is largely understood. Similarly, treatment and prevention strategies are broadly agreed on. Despite this, anaemia in pregnancy remains a common problem in many developing countries. The following strategies could alleviate the factors that contribute to this situation.

8.1 Improve central management

Although oral iron supplementation is a relatively simple intervention, it nonetheless requires a distribution and monitoring infrastructure if it is to be successful.

Although an MCH structure may exist in a country, there is often no explicit commitment to decrease the prevalence of anaemia in pregnancy. For such a programme to be successful, a specific structure of responsibilities needs to be established for management, implementation and evaluation.

8.2 Provide a regular supply of appropriate drugs to antenatal clinics

Financial constraints may make the purchase of a regular supply of drugs difficult. However, the drugs involved are relatively inexpensive and the problem is often one of

poor management in drug procurement. In addition, there may be logistic problems associated with regular transport of drugs to antenatal clinics, especially in rural areas. This may be due to a shortage of fuel or vehicles, but again it is often a symptom of poor central management. There is thus a need for countries to estimate correctly drug requirements in order to allocate the appropriate amount of funds. It was also felt by the Working Group that the drugs required for treatment and supplementation programmes should be integrated into a country's basic drug requirements and not seen as a separate issue. Irregular supplies at the district level may lead to hoarding by providers and to inadequate or sporadic distribution to patients.

8.3 Adequately train antenatal staff

Antenatal clinic staff should understand the basic principles with regard to the prevalence, aetiology, prevention and treatment of anaemia in pregnancy and should be aware of the particular circumstances in their local area which may have an impact on anaemia prevalence. Adequate local training programmes are of vital importance in producing staff with sufficient knowledge and motivation to put local schemes into practice. It is also important that staff, especially in rural areas, be adequately supported in terms of regular supplies of drugs and regular supervisory visits by senior staff. Serious consideration should be given to the involvement of the local health workers, such as the village health worker and the traditional birth attendant. It may be possible to teach them simple techniques for screening for anaemia and to supply them with appropriate drugs, along with adequate education on their use. They could also be involved in local health education efforts and thus increase compliance. This strategy for distribution of supplements requires considerable monitoring and ongoing support on the part of health services.

8.4 Increase compliance

Supplementation programmes may be relatively unsuccessful because of poor compliance by pregnant women (see section 7.1). This may be associated with poor understanding of the need to take tablets, especially if women are feeling relatively well. Tablets may also be passed on to others, rather than taken by women themselves, either for financial gain or because other family members may appear ill. Side-effects of the drugs may also reduce compliance. This highlights the importance of the antenatal clinic for continued health education, a role which should be emphasized to the antenatal staff. Community health workers and, in particular, TBAs, also have an important role in health education. A much bigger promotional effort is required for iron supplementation, similar in scale to that undertaken for immunization. The 10 most important MCH messages (including iron) should be incorporated into general health promotion activities. There is a need for posters, slogans, TV and/or radio messages.

8.5 Establish contact with women who do not attend antenatal clinics

Pregnant women, particularly in poorly-served rural areas, may not attend an antenatal clinic, either because it is too far from their homes, or because they do not understand the importance of early and regular attendance. The health services should try to establish contact with such women through local women's groups, schools, local leaders, etc. A relatively untapped source of support could be the men in the community, who may be instrumental in persuading their wives, sisters, daughters to seek and take iron supplements. A community education programme in Malawi, for example, found that bringing together groups of men and women achieved great and unexpected success.

Another possibility is to use schoolgirls to transmit health messages to other girls not in school.

An alternative way of reaching pregnant women is the use of mobile teams to take antenatal services out to the community. These teams are dependant on appropriate and regular transport and should be able to conduct simple laboratory screening tests.

8.6 Conduct proper programme assessments

It is important to be able to ascertain the extent to which a supplementation programme is succeeding in decreasing the prevalence of anaemia in pregnancy. Weak links should be identified and modifications implemented. Proper programme assessments cannot be made unless current, local information is available on prevalence and aetiology of anaemia in pregnancy. Where this is lacking, active steps should be taken to collect such data, preferably prior to the introduction of a preventive programme.

8.7 Set realistic goals

It is important that realistic goals be set in efforts to decrease the prevalence of anaemia in pregnancy. All people involved in such programmes should have goals. Maintaining motivation in this way can be the critical factor in determining whether a programme has only limited or a lasting effect.

8.8 Look for anaemia-causing factors in antenatal clinics

Failure to control pregnancy-related anaemia may also be associated with inadequate control of factors that contribute to anaemia. This may include inadequate nutritional or family planning programmes or failure of programmes such as malaria or hookworm control.

9. GUIDELINES ON THE DETECTION, PREVENTION AND MANAGEMENT OF ANAEMIA IN PREGNANCY

In this report, and in accordance with WHO nomenclature, three levels of health care are considered:

(i) Community level: Care at this level is provided by individuals working and living within the community and operating, in most cases, outside the formal health care system. These persons - village health workers, traditional birth attendants, traditional healers - usually provide the first contact between the individual and the health care system.

(ii) Health centre level (primary level): This level comprises the most basic level of the formal health care system. Care at this level is provided by trained health workers in a fixed place where essential drugs and basic curative and preventative health services are available. Laboratory facilities are usually absent, but there may be trained staff to do simple microscopy, haemoglobinometry, or basic screening tests.

(iii) District hospital level (first-referral level): Care at this level is provided by trained health staff, including doctors. Drugs, equipment, emergency back-up services are available as well as laboratory facilities to perform routine laboratory tests and inpatient

services are provided. The district hospital would be the place where women at high risk are referred prenatally or sent for emergency obstetric care.

9.1 Detection of anaemia at community level

Routine invasive screening tests are usually inappropriate at the community level, therefore clinical screening methods (qualitative assessment) are recommended.

At the community level, health workers should perform the following two steps in order to identify pregnant women who should be referred to a higher level of care:

9.1.1 Take an obstetric history by asking the pregnant woman the following questions:

(a) Has she been feeling tired or breathless after doing a normal day's work?

(b) Has she experienced significant vaginal bleeding in this pregnancy (more than a cup full)?

(c) Has she delivered a child in the last 12 months?

(d) Is she a young primigravida?

(e) Did she experience severe bleeding in her last pregnancy, and as a consequence was she weak and tired during the postpartum period?

(f) Did she require a blood transfusion for her last delivery?

(h) Did she have a retained placenta necessitating manual removal during the last pregnancy?

A positive history for any one of these factors should alert the health worker and raise the suspicion of anaemia.

9.1.2 The health workers should examine all women for pallor of the conjunctivae, tongue, palms and nail beds using anaemia recognition cards if available.

If on examination the woman looks pale, she should be referred to the nearest health centre where there are facilities for simple laboratory screening tests to confirm the presence of anaemia and possibly for assessing the degree of anaemia (quantitative), as well as facilities to look at a thick and thin film (microscopy).

9.2 Prevention and treatment of anaemia at the community level

9.2.1 Prevention

If there are no high risk factors and the woman is not pale on examination, she should be given standard supplementation as follows:

(a) A combination tablet (60 mg iron + 250 µg folic acid) twice a day, or a ferrous sulphate tablet (60 mg of elemental iron) twice a day and a folic acid tablet daily (1 mg), to be taken throughout pregnancy and the lactation period.

(b) Malaria chemoprophylaxis, depending on the local situation and national policy. When the woman returns for her usual antenatal check ups, she should be reexamined for paleness and it should be screened in the same manner as at the community level.

(c) Anthelminthics if necessary, but not in the first trimester.

(d) Advice on diet: iron rich foods (red meat, dark green leaves, pulses) should be eaten with some animal foods or fresh fruit or tuber staples. Tea or coffee with the meal should be avoided.

9.2.2 Treatment

All women identified as anaemic and referred to a higher level of care should begin the following treatment immediately:

(a) A combination tablet (60 mg iron + 250 µg folic acid) three times a day, or a ferrous sulphate tablet (60 mg of elemental iron) three times a day and a folic acid tablet (1 mg) daily, to be taken for the duration of pregnancy.

(b) Malaria chemoprophylaxis, depending on the local situation and the national policy. In case of chloroquine resistance, other drugs should be used.

(c) An appropriate anthelminthic, such as albendazole (400 mg) after the first trimester, if she is in a hookworm endemic area.

An attempt should also be made to see the woman in one week to check whether she went to the health centre and is taking her medication. All action and advice should be recorded on the woman's health record.

9.3 Detection of anaemia at health centre level

Women whose first contact is at the health centre should be screened in the same manner as at the community level.

For women identified as anaemic or who have been referred from the community level, a screening test with copper sulphate solution or other simple method should be performed.

If possible, women with a positive qualitative screening test (below cut-off value), should have a haemoglobin concentration determination. Referral may be advisable, if circumstances permit, to perform either a haemoglobin or a haematocrit measurement by using a simple haemoglobin comparator (AO Spencer/BMS) or a haematocrit centrifuge.

Based on these measurements and physical findings, anaemia can be categorized as follows:

Degree of anaemia	Hb (g/l)	PCV (%)
Moderate	70-109	24-37
Severe, asymptomatic, less than 34 weeks of gestation	40-69	13-23
Severe, asymptomatic, 34 weeks or more of gestation	40-69	13-23
Severe, symptomatic, irrespective of gestational age	40-69	13-23
Very severe	<40	<13

The woman is considered "symptomatic" if she reports symptoms such as easy fatigability and breathlessness on mild exertion, weakness, or dizziness.

If a microscope is available, additional useful information will be gained by examining a thin blood film for red cell size and colour, and a thick blood film for malaria parasites.

9.4 Prevention and treatment of anaemia at the health centre level

9.4.1 Prevention

If the woman regularly attends the clinic, if there are no risk factors in her history as defined above, and if she does not appear pale on clinical examination, she should receive the standard oral iron and folate supplementation as described above (see 9.2.2).

If she has a high risk factor, appears pale, or has been referred from the community level and the screening test suggests that she is not anaemic, she should be treated as in 9.2.2 above and referred back to the community.

9.4.2 Treatment

Pregnant women with a haemoglobin concentration <110 g/l (or less than 105 g/l in the second trimester) and living in a malarious area

- All women with a haemoglobin below 110 g/l and living in or visiting malaria-endemic areas should have a thick blood film examined. If the slide is positive for malaria, a curative anti-malarial regime should be prescribed. Women who still have positive blood films after having been treated prior to referral, should be given a course of the second line anti-malarial according to the local and national situation.

- Iron/folate combination tablet should be given three times a day, or ferrous sulphate (containing 60 mg elemental iron) three times a day along with folate (1 mg) once daily.

- If hookworm is endemic in the region, an appropriate anthelminthic, if not already taken, should be given.

If the haemoglobin concentration is 70-109 g/l (moderate anaemia)

- Iron and folate treatment should be started as stated above, and the woman seen again in 1 month.

- If at the 1 month visit, the haemoglobin has increased by at least 20 g/l or reached a level >110 g/l (or >105 g/l in the second trimester), treatment should be continued as stated above.

- If the haemoglobin has not increased by at least 20 g/l, this could be due to either:

  genuine treatment failure either because the anaemia is not due to iron/folate deficiency, or there is poor absorption of the supplements. A thin blood film should be examined if facilities are available. If the picture shows microcytic hypochromic anaemia, a course of intramuscular iron or, if facilities permit and staff with adequate skills are available, a total dose infusion (TDI) of iron could be given. If the blood film does not reveal microcytic hypochromic anaemia, 10 ml of intramuscular iron should still be given and the woman referred to the next level for further investigation of her anaemia. Thalassaemia, which also presents with a microcytic hypochromic anaemia, should be ruled out wherever appropriate.

or poor compliance.

- If there are no facilities to examine a thin film, the woman should be referred to the next level of care after being given 10 ml of intramuscular iron.

If the haemoglobin concentration is 40-69 g/l (severe anaemia), the woman is asymptomatic and less than 34 weeks pregnant

- If the blood picture is not microcytic, she should be referred to the next level for further investigation of the cause of her anaemia. (Some of the likely diagnoses include folate deficiency, sickling disorders, AIDS.)

- If the picture is microcytic, the anaemia is likely to be due to iron deficiency and further management depends on whether the period of gestation is greater or less than 28 weeks (this cut-off date ensures that there is enough time for oral supplementation to be successful):

  If the woman is less than 28 weeks pregnant, she should continue on oral supplementation. She should be followed up regularly to ensure that her haemoglobin concentration is rising.

  If the woman is more than 28 weeks pregnant, she should receive a course of intramuscular iron or, if possible, TDI of iron. She should also continue with oral folic acid (1 mg daily).

- If the blood film is not conclusive, or if the region has a significant prevalence of thalassaemia, it is justifiable to give 10 ml of intramuscular iron because of a possible co-existing iron deficiency, and follow up in 2 weeks. If there is evidence of a response, treatment can continue; if not, the woman should be referred to the next level of care.

If the haemoglobin concentration is 40-69 g/l (severe anaemia) and the woman is asymptomatic and more than 34 weeks pregnant

It is possible that the woman may go into preterm labour. Therefore treatment should be initiated with 10 ml of intramuscular iron, and the woman referred to the next level, i.e., the district hospital.

If the haemoglobin concentration is 40-69 g/l and the woman is symptomatic, or if the haemoglobin concentration is less than 40 g/l, regardless of gestational age

This is a medical emergency. Such women are in or likely to develop cardiac failure. Arrangements should be made for immediate transfer to the district hospital.

9.5 Diagnosis and treatment of anaemia at district hospital level

9.5.1 Diagnosis of the cause of anaemia

Anaemic pregnant women who do not respond to treatment at the health centre level should be referred to the district hospital. This is essential where facilities for blood film examination and a haemoglobinometer or a haematocrit centrifuge are not available at the health centre level in order to determine the haemoglobin concentration level or haematocrit and to diagnose the cause of anaemia.

In cases where the thin blood film is normal or doubtful, further laboratory tests such as a bone marrow aspirate or a test for HIV antibody will be necessary.

9.5.2 Treatment

Referral to the district hospital will also be necessary for severely anaemic patients who are more than 34 weeks pregnant. Treatment will consist of TDI, or in very severe cases, blood transfusion. However, as treatment of severe anaemia with whole blood can cause congestive heart failure, it is preferable to use, where available, packed cells together with a rapidly acting diuretic such as furosemide or ethacrynic acid. The woman should stay in the hospital or in a maternity waiting home until delivery.

Referral to the district hospital will also be necessary for symptomatic pregnant women with severe anaemia regardless of gestational age, as such women are in or likely to develop cardiac failure. Management of these cases will consist in transfusion of packed cells under diuretic cover, and delivery if appropriate.

10. RECOMMENDATIONS FOR RESEARCH

10.1 Epidemiological research

10.1.1 Prevalence studies

There are wide regional variations in the prevalence of anaemia. Where up-to-date information is not available, simple prevalence studies should be undertaken. This would ensure relevant, contemporary and accurate baseline data against which any new intervention strategies could be judged.

During the African Regional Consultation on Control of Anaemia in Pregnancy, a model for such a study was proposed.[12] Prevalence studies should entail the obtaining of simple demographic data from pregnant women at their first antenatal clinic attendance such as age, social class, education, parity, number of previous abortions, stillbirths and living children, period of gestation and date of attendance. In addition, a measurement of the Hb concentration or a packed cell volume should be made. This study not only provides prevalence data, but may also identify groups at special risk.

10.1.2 Aetiology studies

The extent to which various aetiological factors operate in different areas may influence the prophylactic and therapeutic measures to be undertaken. For example, hookworm infection may play a major role in iron deficiency in some areas, but a negligible one in others.

Aetiology studies are designed to establish the causes of severe anaemia (Hb <70 g/l). A model for such a study is a case-control study where at least 30 consecutive pregnant women admitted to hospital with haemoglobin levels below 70 g/l are compared with a similar number of matched controls with haemoglobin levels above 110 g/l with respect to reticulocyte count, thin and thick blood films, Hb electrophoresis, stool and urine examination, anti-HIV serology and a bone marrow examination. Thus the frequency and relative contributions of iron and folate deficiency, malaria, haemoglobinopathies, HIV infection, hookworm and schistosomiasis can be calculated.

10.2 Technological research and development

10.2.1 Sensitivity and specificity studies

Studies comparing the sensitivity and specificity of different screening methods for anaemia are needed. For example, a comparison of the sensitivity and specificity of clinical screening for anaemia against an invasive method such as the copper sulphate method could be undertaken using colorimetric Hb determination or the haematocrit as the reference.

10.2.2 Screening methods

In order to improve accuracy and diagnosis, there is a need for refinement of the Tallqvist technique. The filter paper and the standard colour scale could be improved. At present the standard scale ranges from 30% to 100% in units of 10%. It may be possible to subdivide it more simply into three degrees corresponding to <30%, 30-70% and >70% to facilitate colour matching.

There is considerable scope for improvement of anaemia recognition cards for assessing conjunctival or tongue colour.

Feasibility studies are needed on possible introduction of invasive screening tests (Tallqvist and copper sulphate) at the community level.

10.2.3 Development of non-dilutional techniques for Hb determination

Because the accuracy of dilutional techniques is adversely affected by manual dilution, there is a need for further development of accurate and reliable non-dilutional techniques.

The introduction of solar power for battery-operated haemoglobinometers should be further explored.

10.3 Health services research

In order to address the issues of why the prevalence of anaemia remains high although the means to prevent it are known, and to examine the reasons why countries have not been able to implement measures effectively, there is a need for both health systems and operational research on the effectiveness of different interventions.

10.3.1 Distribution systems

Surveys of existing local drug supply systems need to be conducted to determine their effectiveness. This would imply training at the periphery of the health system to better estimate drug requirements and at the central level on allocation of funds. Based on the findings of such surveys, new local supply systems could be developed and their effectiveness evaluated.

10.3.2 Packaging

Local initiatives with regard to low cost packaging of drugs need to be evaluated.

The meeting recommended that UNICEF consider packaging the combined iron/folate tablet in containers of 90 or 100 tablets to make local distribution easier.

10.3.3 Health education

Surveys on the knowledge, attitudes and practices of women, husbands, community leaders, etc, with regard to health during pregnancy and the taking of supplements need to be conducted before planning health education programmes and developing relevant messages.

There is a major gap in research on the effects of the attitudes of health care providers on adherence to supplementation programmes. Operational research on the education and supervision of health care personnel is needed.

10.3.4 Communication

Local production of culturally appropriate messages to help promote the increased use of supplementation and of iron/folate rich foods during pregnancy should be encouraged and the effectiveness of the messages evaluated. Culturally appropriate teaching methods such as songs or plays on the radio and TV, or at local festivals, need also to be developed and evaluated.

10.3.5 Compliance

Non-compliance remains a major problem in most anaemia control programmes. The factors that influence women's compliance and health care utilisation need to be examined in order to develop and implement successful antenatal services and supplementation programmes. In particular, levels of compliance need to be evaluated in relation to accessibility and acceptability of antenatal services and availability of drugs at any given time.

There is a need for cross-cultural surveys to examine women's attitudes to taking supplements during pregnancy or seeking health care services, their perceptions of their pregnancy state, local beliefs and practices regarding health during pregnancy, and other factors influencing women's health decision-making.

With such information, antenatal services and, in particular, anaemia control programmes, may be adapted to be more culturally appropriate. For example, iron/folate tablets that are more acceptable in terms of colour, taste and smell could be produced.

Studies are required to elicit information on the efficacy of using different categories of health personnel for medication distribution, e.g. traditional birth attendants and village health workers, and the training and motivation required to sustain the health workers' commitment to the programme.

10.4 Pharmacological studies

10.4.1 Alternative drug delivery systems

Research is under way to develop iron preparations that are better absorbed, without increasing the frequency of side-effects or raising the cost. Research on alternative preparations that need to be taken only once daily has indicated that such treatment regimes may improve bioavailability of iron, reduce side-effects and increase compliance.

10.4.2 Intramuscular iron

In areas of high anaemia prevalence where compliance with oral supplementation is poor, studies are needed to evaluate the acceptability of, adverse reactions to, and effect on iron status of intramuscular iron.

10.5 Clinical trials

Whether routine iron supplementation causes any harm in well nourished communities is still unclear and will not be clarified without further large-scale randomized trials.[55]

There are few data on the effects of iron supplementation in communities where iron deficiency is common and anaemia is a serious clinical problem. Trials are needed in these populations, particularly looking at the effects of iron supplementation on substantive outcomes as well as adverse effects.

The existing evidence suggests that routine folate supplementation may reduce the risk of preterm delivery and low birth weight and that combined with malaria

chemoprophylaxis, it may have beneficial effects on other important outcomes of pregnancy. Well-conducted randomized trials will establish this with confidence.

11. CONCLUSION

Anaemia in pregnancy remains widely prevalent in developing countries and is an important contributor to maternal mortality and morbidity. The aetiology of anaemia is largely understood and treatment and prevention strategies are broadly agreed on. Nonetheless, anaemia remains a serious problem for many reasons, including poor central management, logistic problems, lack of adequate staff training, poor compliance and inaccessibility of antenatal services. Successful anaemia control programmes require a specific commitment by countries to decrease the prevalence of anaemia in pregnancy. Improved organization of the primary health care system, improved antenatal care services and regular supply of drugs are all needed. The Working Group recommended that training programmes for antenatal staff be developed with more emphasis on anaemia prevention, the involvement of traditional health workers such as TBAs in anaemia control programmes, and better promotional efforts for iron supplementation.

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ANNEX 1

MATERNAL HEALTH AND SAFE MOTHERHOOD
Division of Family Health

Meeting of Technical Working Group on
Prevention and Treatment of Severe Anaemia in Pregnancy

Geneva, 20-22 May 1991

PROVISIONAL AGENDA AND TIMETABLE

Monday 20 May

9:00-10:30	Opening and welcome. Introduction of participants. The WHO Maternal Health and Safe Motherhood Research programme. Selection of chairperson and rapporteur. Introduction of agenda and objectives. Activities to date.
10:30-10:45	Coffee break
10:50-12:00	Agenda item 4: Identification of shortcomings and constraints in current modes of prevention of severe pregnancy anaemia. Why have so many anaemia prevention programmes not been successful? 4.1 Screening, making contact with pregnant women. Feasibility of prepregnancy interventions. 4.2 Prenatal care, its coverage, content and quality. Diagnosis and treatment of nutritional and non-nutritional causes of anaemia. Iron supplementation and other treatment regimes. Ongoing monitoring. 4.3 Technological issues: haemoglobinometry, standards,drug formulations and modes of delivery, drug availability, supplies costs, etc. 4.4 Compliance, health education.
12:00-14:00	Lunch
14:00-15:30	Agenda item 4 continued
15:30-15:45	Tea break
15:50-17:00	Agenda item 4 continued

Tuesday 21 May

9:00-10:30 Agenda item 5: Treatment of severe pregnancy anaemia.

5.1 Definitions of severe anaemia, taking into account gestational age. Hb cut-off points.
5.2 Treatment at primary health care level. Feasibility, risks and benefits of alternative forms of treatment. Criteria for transfer.
5.3 Treatment at referral centre. Feasibility, risks and benefits of alternative forms of treatment.
5.4 Technological and logistic issues, e.g. blood supplies.

10:30-10:45 Coffee break

10:50-12:00 Agenda item 5 continued

12:00-14:00 Lunch

14:00-15:30 Agenda item 5 continued

15:30-15:45 Tea break

15:50-17:00 Agenda item 6: Proposed Guidelines:

6.1 Intended readership
6.2 Topics to be covered of those discussed under items 4 and 5.
6.3 Additional topics, including adaptation to local cultural and health service settings (beliefs as well as coverage of prenatal care), checklists, how to assess the local situation, identifying obstacles, etc.

Wednesday 22 May

9:00-10:30 Agenda item 7: Operational research needs:

7.1 Identification of topics arising from preceding discussions needing further clarification.
7.2 Review of proposals submitted to WHO's Maternal Health and Safe Motherhood research programme.

10:30-10:45 Coffee break

10:50-12:00 Agenda item 7 continued

12:00-14:00 Lunch

14:00-15:30 Agenda item 8: Any other business, including recommendations to WHO on follow up action.

15:30-15:45 Tea break

15:50- Closure

ANNEX 2

MATERNAL HEALTH AND SAFE MOTHERHOOD
Division of Family Health

Meeting of Technical Working Group on
Prevention and Treatment of Severe Anaemia in Pregnancy

Geneva, 20-22 May 1991

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ANNEX 3

RESEARCH PROPOSALS CONSIDERED BY THE WORKING GROUP

Study on anaemia in pregnancy, Ratnam et al.

This multicentre study was to compare the existing oral iron supplementation programme in pregnancy with a parenteral one. Oral iron would be replaced by two intramuscular doses of 500 mg of elemental iron given 4 weeks apart. Both groups would receive, in addition to iron, oral folate and an oral anthelminthic.

Initially, the Nutrition Unit had expressed reservations about the use of parenteral iron, but the Unit's representatives at the meeting felt that as there was reasonable medical justification they would have no further objections. None of any other members of the group had any reservations about giving parenteral iron as suggested in the trial protocol. However, as intramuscular iron injections can be painful, it was suggested that some attempt should be made to gather data on what the women thought about it.

It was also felt that too many endpoints were being measured and it was suggested that the study concentrate on the following:

- Hb increase
- Blood transfusion requirements
- Birth weight
- Perinatal deaths

Thus the general feeling of the group was to recommend support for this study, but that funding may be available for only a proportion of the centres to be involved in the study.

An epidemiological study on blood transfusion, Ogunsanwo et al.

The aim of this study was to examine the current transfusion practices in a teaching hospital in Nigeria as they relate to obstetric patients, with a view to improving the appropriate use of blood and to promote similar initiatives in other areas in Nigeria. It was proposed that the study be jointly funded by GBSI and SMI. A critique of the original proposal had already been communicated to GBSI and a revised proposal had been drawn up to reflect consideration of these comments. An outside consultant had also visited Nigeria on behalf of GBSI and had come back with a favourable report.

In general it was felt that the revised protocol was acceptable, but there were still some reservations about the amount of data to be collected and the complexity of the data collection forms. It was thus suggested that the forms be reviewed by an expert in this area and revised if necessary.

Reservations were also expressed about the amount of funding required. But the GBSI representatives felt that on careful review this was justified, especially in light of the recent report by the independent consultant.

Research on pregnancy anaemia: strategies to improve compliance with iron supplementation, Seshadri et al.

As a revised protocol had only just been received, the group felt that it had had insufficient time to properly consider it. Although certain methodological changes had been made to take into account previous comments on the original protocol, it was felt that there was insufficient detail given in the revised version, particularly in the budgetary requirements and in the time plan of the work. It was thus suggested that the proposal be considered at a future suitable meeting.